
Math 49

Precalculus with Limits

Math 49 Course at a Glance

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

Review Content	
REVIEW	SECTIONS 1.2, 1.5, 4.1, 4.2, 4.3, 4.4, 4.5, 4.7
LEARNING OBJECTIVES	
M49-FND-001	I can determine the domain and range of a function from a formula, graph, or table.
M49-FND-002	I can describe and apply transformations of a parent function.
M49-FND-004	I can analyze exponential functions using their equations, graphs, and contexts.
M49-FND-005	I can analyze logarithmic functions using their equations, graphs, and contexts.
M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses	
REQUIRED	SECTIONS 1.4, 1.7
LEARNING OBJECTIVES	
M49-FND-003	I can determine whether a function is even, odd, or neither.
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain.
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables.
M49-OPS-003	I can write a formula for the composition of two functions and determine its domain.
M49-OPS-004	I can decompose a function into a composition of simpler functions.
M49-OPS-005	I can determine whether a function has an inverse on a given domain.
M49-OPS-006	I can find and verify an inverse function algebraically.
M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables.

Power, Polynomial, and Rational Functions	
REQUIRED	SECTIONS 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4.2, 4.4
LEARNING OBJECTIVES	
M49-PWR-001	I can identify a power function and interpret the parameters in $y = kx^p$.
M49-PWR-002	I can construct a power function from two data points or a proportional relationship.
M49-PWR-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M49-PWR-004	I can recognize and model direct, inverse, and other power-variation relationships.
M49-PWR-005	I can solve equations involving integer or rational powers.
M49-PWR-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M49-PWR-008	I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
M49-ALG-002	I can predict the end behavior of a polynomial from its degree and leading coefficient.
M49-ALG-003	I can determine polynomial zeros and their multiplicities.
M49-ALG-004	I can construct a polynomial function from specified zeros or graphical features.
M49-ALG-005	I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
M49-ALG-006	I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
M49-ALG-007	I can sketch and analyze a rational function using its algebraic and asymptotic features.
M49-ALG-008	I can build or select a power, polynomial, or rational model for a context.

Limits and Function Comparison	
REQUIRED	SECTIONS 12.1
LEARNING OBJECTIVES	
M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input.
M49-LIM-002	I can estimate a limit from a graph or table.
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification.
M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior.
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior.
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

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Sequences and Series		Additional Limit Concepts		Parametric Equations		Conic Sections	
REQUIRED	SECTIONS 11.1, 11.2, 11.3, 11.4	EXTENSION	SECTIONS 12.1	EXTENSION	SECTIONS 8.6, 8.7	EXTENSION	SECTIONS 10.1, 10.2, 10.3, 10.5
LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES	
M49-SEQ-001	I can evaluate and graph a sequence defined explicitly or recursively.	M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists.	M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.	M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
M49-SEQ-002	I can write explicit and recursive formulas for arithmetic sequences.	M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point.	M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation.	M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form.
M49-SEQ-003	I can write explicit and recursive formulas for geometric sequences.			M49-PAR-003	I can create or interpret parametric equations for motion in the plane.	M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle.
M49-SEQ-004	I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.					M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse.
M49-SEQ-005	I can create and interpret a sequence model for discrete change.					M49-CON-005	I can write, interpret, and graph implicit and parametric equations of a hyperbola.
M49-SER-001	I can interpret and use summation notation.					M49-CON-006	I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
M49-SER-002	I can calculate a finite arithmetic series.					M49-CON-007	I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
M49-SER-003	I can calculate a finite geometric series.					M49-CON-008	I can explain a parabola as the set of points equidistant from a focus and directrix.
M49-SER-004	I can determine whether an infinite geometric series converges and find its sum when it does.						
M49-SER-005	I can create and interpret a series model for accumulated change.						

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-FND-001

I can determine the domain and range of a function from a formula, graph, or table.
TEXTBOOK SECTIONS 1.2 Domain and Range

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
R	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Reinforce
R	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Reinforce

M49-FND-002

I can describe and apply transformations of a parent function.
TEXTBOOK SECTIONS 1.5 Transformation of Functions

SUPPORTING SKILLS

R	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Reinforce
R	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Reinforce
R	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Reinforce

M49-FND-004

I can analyze exponential functions using their equations, graphs, and contexts.
TEXTBOOK SECTIONS 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify horizontal asymptotic behavior of an exponential graph. (first introduced in Math 31) SK-EXP-ASYMPTOTE · Reinforce
R	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-FND-005

I can analyze logarithmic functions using their equations, graphs, and contexts.
TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify vertical asymptotic behavior of a logarithmic graph. (first introduced in Math 31) SK-LOG-ASYMPTOTE · Reinforce
R	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

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SKILL PROGRESSION

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Introduce

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Reinforce

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Deepen

A

Apply

M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties. TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models
SUPPORTING SKILLS	
R	Rewrite exponential expressions using a common base when possible. (first introduced in Math 31) SK-EXP-REWRITE-BASE · Reinforce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
R	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Reinforce
R	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Reinforce

Function Operations, Composition, and Inverses

REQUIRED

UNIT TEXTBOOK COVERAGE

1.4 Composition of Functions; 1.7 Inverse Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-FND-003	I can determine whether a function is even, odd, or neither. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
D	Test a formula or graph for even or odd symmetry. (first introduced in Math 31) SK-FUNC-SYMMETRY · Deepen
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain. TEXTBOOK SECTIONS 1.4 Composition of Functions
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Determine the domain of a sum, difference, product, quotient, or composition. SK-FUNC-DOMAIN-COMBINE · Introduce
I	Add, subtract, multiply, and divide function outputs. SK-FUNC-OPERATIONS · Introduce
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables. TEXTBOOK SECTIONS 1.4 Composition of Functions
SUPPORTING SKILLS	
D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Follow intermediate values to compose functions from tables. (first introduced in Math 31) SK-FUNC-COMPOSE-TABLE · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M49-OPS-003	I can write a formula for the composition of two functions and determine its domain. TEXTBOOK SECTIONS 1.4 Composition of Functions
SUPPORTING SKILLS	
D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Determine the domain of a sum, difference, product, quotient, or composition. (first introduced in Math 39) SK-FUNC-DOMAIN-COMBINE · Deepen
M49-OPS-004	I can decompose a function into a composition of simpler functions. TEXTBOOK SECTIONS 1.4 Composition of Functions
SUPPORTING SKILLS	
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
I	Identify simpler inner and outer functions in a composition. SK-FUNC-DECOMPOSE · Introduce
M49-OPS-005	I can determine whether a function has an inverse on a given domain. TEXTBOOK SECTIONS 1.7 Inverse Functions
SUPPORTING SKILLS	
D	Apply the horizontal line test for invertibility. (first introduced in Math 31) SK-FUNC-INVERSE-HLT · Deepen
R	Test whether a function is one-to-one on a domain. (first introduced in Math 31) SK-FUNC-ONE-TO-ONE · Reinforce

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SKILL PROGRESSION

I

Introduce

R

Reinforce

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Deepen

A

Apply

M49-OPS-006	I can find and verify an inverse function algebraically. TEXTBOOK SECTIONS 1.7 Inverse Functions
SUPPORTING SKILLS	
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
D	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Deepen
D	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Deepen

M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables. TEXTBOOK SECTIONS 1.7 Inverse Functions
SUPPORTING SKILLS	
D	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen
D	Reverse input and output units when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-UNITS · Deepen

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Power, Polynomial, and Rational Functions

REQUIRED

UNIT TEXTBOOK COVERAGE

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions; 3.7 Rational Functions; 3.8 Inverses and Radical Functions; 3.9 Modeling Using Variation; 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-PWR-001

I can identify a power function and interpret the parameters in $y = kx^p$.

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Interpret the coefficient and exponent in a power function. (first introduced in Math 39) SK-POWER-PARAMETER-INTERPRET · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-002

I can construct a power function from two data points or a proportional relationship.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Determine a power function from two points or a proportional relationship. (first introduced in Math 39) SK-POWER-CONSTRUCT · Deepen

M49-PWR-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Determine power-function behavior as the input approaches zero or positive or negative infinity. (first introduced in Math 39) SK-POWER-END-BEHAVIOR · Deepen
D	Relate the coefficient and exponent of a power function to its graph shape. (first introduced in Math 39) SK-POWER-SHAPE · Deepen
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-PWR-004

I can recognize and model direct, inverse, and other power-variation relationships.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

D	Represent and interpret a direct-variation relationship. (first introduced in Math 39) SK-VAR-DIRECT · Deepen
D	Represent and interpret an inverse-variation relationship. (first introduced in Math 39) SK-VAR-INVERSE · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-005

I can solve equations involving integer or rational powers.

TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Deepen

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M49-PWR-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
TEXTBOOK SECTIONS 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A

Compare functions shown in different representations. (first introduced in Math 21)
SK-FUNC-COMPARE-REP · Apply

D

Compare the long-run growth or decay of functions from different families. (first introduced in Math 39)
SK-FUNC-DOMINANCE · Deepen

M49-PWR-008

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A

Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21)
SK-EXP-RATIONAL · Apply

A

Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21)
SK-FUNC-DOMAIN-EVEN-ROOT · Apply

I

Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
SK-POWER-FRACTIONAL-ANALYZE · Introduce

M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.
TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

I

Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · Introduce

A

Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12)
SK-POLY-VOCABULARY · Apply

M49-ALG-003

I can determine polynomial zeros and their multiplicities.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

I

Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · Introduce

D

Determine real and complex zeros using an appropriate method. (first introduced in Math 21)
SK-POLY-ZEROS · Deepen

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D

Build a polynomial from zeros, multiplicities, and a point. (first introduced in Math 31)
SK-POLY-CONSTRUCT · Deepen

A

Relate zero multiplicity to local graph behavior. (first introduced in Math 39)
SK-POLY-MULTIPLICITY · Apply

A

Determine real and complex zeros using an appropriate method. (first introduced in Math 21)
SK-POLY-ZEROS · Apply

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D

Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39)
SK-POLY-END-BEHAVIOR · Deepen

D

Relate zero multiplicity to local graph behavior. (first introduced in Math 39)
SK-POLY-MULTIPLICITY · Deepen

A

Determine real and complex zeros using an appropriate method. (first introduced in Math 21)
SK-POLY-ZEROS · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M49-ALG-006

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

A

Cancel common factors rather than individual terms. (first introduced in Math 21)
SK-RAT-CANCEL-FACTORS · Apply

I

Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · Introduce

A

Factor numerators and denominators completely. (first introduced in Math 21)
SK-RAT-FACTOR · Apply

I

Identify a removable discontinuity and its coordinates.
SK-RAT-HOLE · Introduce

I

Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE · Introduce

M49-ALG-007

I can sketch and analyze a rational function using its algebraic and asymptotic features.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

D

Determine horizontal or slant asymptotic behavior. (first introduced in Math 49)
SK-RAT-END-BEHAVIOR · Deepen

D

Identify a removable discontinuity and its coordinates. (first introduced in Math 49)
SK-RAT-HOLE · Deepen

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

I

Determine the sign of a rational function on intervals.
SK-RAT-SIGN-INTERVAL · Introduce

D

Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49)
SK-RAT-VERTICAL-ASYMPTOTE · Deepen

M49-ALG-008

I can build or select a power, polynomial, or rational model for a context.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

I

Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT · Introduce

D

Explain a model parameter in the language of its context. (first introduced in Math 21)
SK-MODEL-PARAMETER-INTERPRET · Deepen

A

Restrict a model to meaningful contextual inputs. (first introduced in Math 12)
SK-MODEL-VALID-DOMAIN · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Limits and Function Comparison

REQUIRED

UNIT TEXTBOOK COVERAGE

12.1 Finding Limits: Numerical and Graphical Approaches

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input. TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches
SUPPORTING SKILLS	
I	Read and write limit notation with correct approach direction. SK-LIM-NOTATION · Introduce
M49-LIM-002	I can estimate a limit from a graph or table. TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches
SUPPORTING SKILLS	
I	Estimate one-sided and two-sided limits from a graph. SK-LIM-GRAPH · Introduce
I	Estimate a limit from input-output values approaching from both sides. SK-LIM-TABLE · Introduce
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Resolve an indeterminate algebraic form by factoring or rationalizing. SK-LIM-SIMPLIFY · Introduce
I	Test a limit using direct substitution. SK-LIM-SUBSTITUTE · Introduce
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply

M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Interpret unbounded output near a finite input as an infinite limit. SK-LIM-INFINITE · Introduce
D	Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49) SK-RAT-VERTICAL-ASYMPTOTE · Deepen
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Compare dominant terms to determine behavior at infinity. SK-LIM-INFINITY · Introduce
D	Determine horizontal or slant asymptotic behavior. (first introduced in Math 49) SK-RAT-END-BEHAVIOR · Deepen
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
I	Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions. SK-LIM-GROWTH-ORDER · Introduce

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Sequences and Series

REQUIRED

UNIT TEXTBOOK COVERAGE

11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-SEQ-001

I can evaluate and graph a sequence defined explicitly or recursively.
TEXTBOOK SECTIONS 11.1 Sequences and Their Notations

SUPPORTING SKILLS

I	Generate an explicit formula from a sequence pattern. SK-SEQ-EXPLICIT · Introduce
I	Graph a sequence as discrete ordered pairs. SK-SEQ-GRAPH · Introduce
I	Interpret subscripts and evaluate terms of a sequence. SK-SEQ-NOTATION · Introduce
I	Generate a recursive formula with an initial condition. SK-SEQ-RECURSIVE · Introduce

M49-SEQ-002

I can write explicit and recursive formulas for arithmetic sequences.
TEXTBOOK SECTIONS 11.2 Arithmetic Sequences

SUPPORTING SKILLS

I	Recognize constant difference in an arithmetic sequence. SK-SEQ-DIFFERENCE · Introduce
D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-003

I can write explicit and recursive formulas for geometric sequences.
TEXTBOOK SECTIONS 11.3 Geometric Sequences

SUPPORTING SKILLS

D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
I	Recognize constant ratio in a geometric sequence. SK-SEQ-RATIO · Introduce
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-004

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
TEXTBOOK SECTIONS 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
D	Recognize constant difference in an arithmetic sequence. (first introduced in Math 49) SK-SEQ-DIFFERENCE · Deepen
D	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Deepen

M49-SEQ-005

I can create and interpret a sequence model for discrete change.
TEXTBOOK SECTIONS 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Apply
I	Identify initial value and pattern of change in a discrete context. SK-SEQ-MODEL · Introduce
A	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Apply

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Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M49-SER-001

I can interpret and use summation notation.
TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A

Interpret subscripts and evaluate terms of a sequence. (first introduced in Math 49)
SK-SEQ-NOTATION · Apply

I

Expand and evaluate a finite sum written in sigma notation.
SK-SER-SIGMA · Introduce

M49-SER-002

I can calculate a finite arithmetic series.
TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A

Recognize constant difference in an arithmetic sequence. (first introduced in Math 49)
SK-SEQ-DIFFERENCE · Apply

I

Apply an arithmetic-series sum formula.
SK-SER-ARITHMETIC · Introduce

M49-SER-003

I can calculate a finite geometric series.
TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A

Recognize constant ratio in a geometric sequence. (first introduced in Math 49)
SK-SEQ-RATIO · Apply

I

Apply a finite geometric-series sum formula.
SK-SER-GEOMETRIC-FINITE · Introduce

M49-SER-004

I can determine whether an infinite geometric series converges and find its sum when it does.
TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A

Recognize constant ratio in a geometric sequence. (first introduced in Math 49)
SK-SEQ-RATIO · Apply

I

Test convergence and sum an infinite geometric series.
SK-SER-GEOMETRIC-INFINITE · Introduce

M49-SER-005

I can create and interpret a series model for accumulated change.
TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A

Explain a model parameter in the language of its context. (first introduced in Math 21)
SK-MODEL-PARAMETER-INTERPRET · Apply

A

Apply an arithmetic-series sum formula. (first introduced in Math 49)
SK-SER-ARITHMETIC · Apply

A

Apply a finite geometric-series sum formula. (first introduced in Math 49)
SK-SER-GEOMETRIC-FINITE · Apply

A

Test convergence and sum an infinite geometric series. (first introduced in Math 49)
SK-SER-GEOMETRIC-INFINITE · Apply

I

Identify the initial term, change factor, and number of terms in an application.
SK-SER-MODEL · Introduce

Math 49 In-Depth Curriculum

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Additional Limit Concepts

EXTENSION

UNIT TEXTBOOK COVERAGE

12.1 Finding Limits: Numerical and Graphical Approaches

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists. TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches
SUPPORTING SKILLS	
D	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Deepen
D	Read and write limit notation with correct approach direction. (first introduced in Math 49) SK-LIM-NOTATION · Deepen
A	Estimate a limit from input-output values approaching from both sides. (first introduced in Math 49) SK-LIM-TABLE · Apply

M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point. TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches
SUPPORTING SKILLS	
I	Check the three conditions for continuity at a point. SK-LIM-CONTINUITY · Introduce
A	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Apply
D	Test a limit using direct substitution. (first introduced in Math 49) SK-LIM-SUBSTITUTE · Deepen

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Parametric Equations

EXTENSION

UNIT TEXTBOOK COVERAGE

8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation. TEXTBOOK SECTIONS 8.7 Parametric Equations: Graphs
SUPPORTING SKILLS	
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Deepen
D	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Deepen

M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation. TEXTBOOK SECTIONS 8.6 Parametric Equations
SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Eliminate a parameter algebraically. (first introduced in Math 32) SK-PAR-ELIMINATE · Deepen
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

M49-PAR-003	I can create or interpret parametric equations for motion in the plane. TEXTBOOK SECTIONS 8.6 Parametric Equations; 8.7 Parametric Equations: Graphs
SUPPORTING SKILLS	
D	Interpret or construct parametric equations for planar motion. (first introduced in Math 32) SK-PAR-MODEL · Deepen
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 49 In-Depth Curriculum

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Conic Sections

EXTENSION

UNIT TEXTBOOK COVERAGE

10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship. TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates
SUPPORTING SKILLS	
I	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. SK-CONIC-CLASSIFY · Introduce
I	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. SK-CONIC-DISTANCE-DEFINITION · Introduce
M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form. TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Complete squares to convert a general conic equation to standard form. SK-CONIC-GENERAL-STANDARD · Introduce
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen

M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Connect implicit, standard, and parametric equations of a circle. SK-CONIC-CIRCLE-EQUATIONS · Introduce
I	Determine and use a circle's center and radius. SK-CONIC-CIRCLE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply
M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse. TEXTBOOK SECTIONS 10.1 The Ellipse
SUPPORTING SKILLS	
I	Connect implicit, standard, and parametric equations of an ellipse. SK-CONIC-ELLIPSE-EQUATIONS · Introduce
I	Determine and use an ellipse's center, axes, vertices, and foci. SK-CONIC-ELLIPSE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M49-CON-005

I can write, interpret, and graph implicit and parametric equations of a hyperbola.
TEXTBOOK SECTIONS 10.2 The Hyperbola

SUPPORTING SKILLS

I	Connect implicit, standard, and parametric equations of a hyperbola. SK-CONIC-HYPERBOLA-EQUATIONS · Introduce
I	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. SK-CONIC-HYPERBOLA-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

M49-CON-006

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Determine and use a circle's center and radius. (first introduced in Math 49) SK-CONIC-CIRCLE-FEATURES · Deepen
A	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. (first introduced in Math 49) SK-CONIC-CLASSIFY · Apply
D	Determine and use an ellipse's center, axes, vertices, and foci. (first introduced in Math 49) SK-CONIC-ELLIPSE-FEATURES · Deepen
D	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. (first introduced in Math 49) SK-CONIC-HYPERBOLA-FEATURES · Deepen

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. (first introduced in Math 49) SK-CONIC-DISTANCE-DEFINITION · Deepen
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.
TEXTBOOK SECTIONS 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

I	Explain a parabola using its focus-directrix distance relationship. SK-CONIC-PARABOLA-DEFINITION · Introduce
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply